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SUPERSEDING
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MILITARY SPECIFICATION

TRAY ASSEMBLY, BATTERY CHARGING MX-10154 ( )/U

This specification is approved for use by the Electronics Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of battery charging tray designated Tray Assembly, Battery Charging MX-10154 ( )/U for furnishing a means of charging from one through six sealed nickel cadmium batteries from a constant current source. (see 6.1)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to US Army Electronics Research and Development Command, ATTN: DELET-R-S, Fort Monmouth, N.J. 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

L-P-393	Plastic Molding Material, Polycarbonate, Injection and Extrusion

MILITARY

MIL-P-116	Preservation, Methods of
MIL-C-3432	Cables, (Power and Special Purpose) and Wire, Electrical (300 and 600 Volts)
MIL-C-3885	Cable Assemblies and Cord Assemblies, Electrical
MIL-S-8805/3	Swith, Push, 10 Amperes and Low-Level, Dusttight
MIL-P-11268	Parts, Materials, and Processes used in Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-I-23053/5	Insulation Sleeving, Electrical, Heat Shrinkable, Polyolefin, Flexible, Crosslinked
MIL-C-39012/21	Connector, Coaxial, Radio Frequency (Series Bnc)Uncabled (Receptables, Female, Jam Nut Mounted, Class 1i)
MIL-E-55585	Electronics Equipment and Parts, Packaging of

STANDARDS

FEDERAL

FED-STD-595	Color (Requirements For Individual Color Chips)

MILITARY

MIL-STD-105	Sampling Procedures and Table For Inspection By Attributes

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MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-726	Packaging Requirements Code
MIL-STD-810	Environmental Test Methods
MIL-STD-1344	Test Methods, For Electrical Connectors
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities
MIL-STD-45662	Calibration Systems Requirements

HANDBOOKS

MILITARY

MIL-HDBK-217	Reliability Prediction of Electronic Equipment

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

US Army Electronics Research and Development Command (ERADCOM)

C5072837 Battery Charging Tray Assembly MX-10154/U

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specified procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Unit Definition. The tray assembly is designed to permit the charging of from 1 to 6 each sealed nickel cadmium batteries from a constant current source.

3.2 First article. When specified, a sample shall be subjected to first article inspection (see 4.4).

3.3 Design and construction. (See 4.3) The unit shall comply with the design and construction requirements of MIL-P-11268 and as specified herein.

3.3.1 Physical characteristics.

3.3.1.1 Dimensions. Length, width, and depth shall meet the outline dimensions shown in ERADCOM Drawing C5072837.

3.3.1.2 Weight. The overall weight of the unit shall not exceed 3.0 lbs. (1.4 kgs)

3.3.2 Mechanical characteristics.

3.3.2.1 Cable. (See 4.3) Two (2) each No. 20 AWG Conductor, medium duty, general purpose, extra flexible, CO-02MGE (2/20) in accordance with MIL-C-3432.

3.3.2.1.1 Cable assembly. The cable assembly shall be in accordance with MIL-C-3885.

3.3.2.2 Battery holding tray. (See 4.2) The tray shall be machined or molded from Lexan plastic material (polycarbonate fiberglass) in accordance with L-P-393.

3.3.2.3 Switches. Two (2) circuit, black button, water-tight 10 ampere, momentary pushbutton switches shall be provided on five (5) of the battery holding compartments. The switches shall be in accordance with MIL-S-8805/3.

3.3.2.4 Battery connectors. (See 4.3) The battery connectors shall be BNC type UG625B/U in accordance with MIL-C-39012/21 but modified per drawing C5072839.

3.3.2.5 Banana plugs. (See 4.3) The cable shall be terminated at one end in a Y configuration with a Red banana plug at POSITIVE and a Black banana plug at NEGATIVE. The Y termination shall employ heat shrink tubing in accordance with MIL-I-23053/5.

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3.3.2.6 Cable clamp. A cable clamp shall be used to secure the cable to the battery tray.

3.3.2.7 Overall Design. The overall design shall include the following:

a. Provision shall be made for the cable to be secured to the battery holding tray for storage.

b. Provisions shall be made for battery trays to be capable of being stacked for storage.

c. Provision shall be made to allow access to the switches, connectors and wiring for maintenance.

3.4 Performance. (See 4.7)

3.4.1 DC Resistance. (See Drawing C5072833) With a jumper cable or shorted connector placed between terminal ends B and C, conductor shall be continuous between terminal ends A and D with a dc resistance of not greater than 0.150 ohms.

3.4.2 Continuity. (See Drawing C5072833) Each conductor in the assembly shall be continuous between terminal ends A to B and C to D.

3.4.3 Insulation resistance. The insulation resistance shall not be less than 100 megohms - 1000 feet.

3.4.4 Dielectric withstanding voltage. There shall be no evidence of breakdown when the cable assemblies are subjected to the test of 4.7.4.

3.4.5 Switch Function. A combination of the weight of the battery plus the insertion force on the BNC connector shall establish contact at the connector while opening the pushbutton switch.

3.5 Environmental conditions. (See 4.8)

3.5.1 Temperature. The unit shall not be adversely affected by the effects of ambient air temperature as follows:

a. Operating: The unit shall meet the performance requirements specified when exposed to the following operating environments:

(1) Normal operation: -25°F(-32°C) to +145°F(63°C).
(2) After storage: Up to one hour operation at +145°F(63°C

b. Non-operating storage and transport: -70°F(-57°C) to +160°F(71°C).

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3.5.1.1 Low temperature flexibility, cable assembly only. (See 4.8.1.3) Each portion of a cable assembly (each breakout in a cable jacket) containing a molded or potted splice, crotch, or other molded or potted section of heat-shrunken transition, shall be tested for low-temperature flexibility. The conditioning and test temperature shall be the lower limit of the cable operating temperature range. If not specified, the temperature shall be -40°F(-40°C). Upon examination after test, there shall be no visible cracks or other damage in the molded section or adjacent portion of the cable jacket, and the assembly shall meet the requirements of 3.4.2, 3.4.3, and 3.4.4.

3.5.2 Humidity. The unit shall meet the performance requirements of 3.4.1 through 3.4.5 after subjected to test of 4.8.2.

3.5.3 Dust. The unit shall meet the performance requirements of 3.4.1 through 3.4.5 after subjected to test of 4.8.3.

3.5.4 Fungus. All materials shall be nonnutrient to fungus growth. The manufacturer shall certify that all materials are fungus resistant (see 4.8.4) or shall perform the test specified in 4.8.4.

3.5.5 Vibration. (See 4.8.5) The unit shall withstand vibration induced during field transport by military vehicle and by rough handling. The unit shall withstand vibrations encountered during transportation and servicing.

3.5.6 Shock. (See 4.8.6) The unit shall withstand shock induced during field transport by military vehicle and by rough handling. The unit shall withstand shock encountered during servicing.

3.5.7 Bench handling shock. The non-operating unit shall not be adversely affected by normal bench handling type shocks, and shall be capable of meeting the performance requirements specified herein after being subjected to the tests of paragraph 4.8.7.

3.5.8 Salt-fog. (See 4.8.8) The unit shall be resistant to the corrosive effects of salt-sea atmosphere.

3.5.9 Moisture resistance. After being subjected to the test of 4.8.9, the unit shall show no visual evidence of blistering, corrosion, or exposure of the cable conductors. The cable assembly shall meet the requirements of paragraph 3.4.3.

3.5.10 Pull test. (See 4.8.10) The terminals and taper pins shall not pull out of the conductor or assembly when subjected to a 10 pound pull. After twisting and pulling, there shall be no more than 1/32 inch slippage of the cable jacket from the connector, nor

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separation of the bond on molded terminations, molded bend reliefs, and molded strain reliefs. Following the test the cable assembly shall meet the requirements of 3.4.1 and 3.4.3.

3.6 Life.

3.6.1 Storage life. The unit and associated component parts, when properly prepared for storage shall have a shelf life of no less than 5 years without refurbishment, extendable to 15 years when refurbished at no less than 5 year intervals.

3.6.2 Operating life. The unit shall have an operating life of no less than 10 years with periodic maintenance.

3.6.3 Total useful life. The unit shall have a minimum total useful life (storage plus operating life) of no less than 15 years.

3.7 Materials, parts, and processes. Materials, parts, and processes used in the unit shall be in accordance with MIL-P-11268. The order of precedence for selection of parts shall be as specified in MIL-STD-143.

3.7.1 Protective coating. Protective coatings of the unit shall be in accordance with MIL-P-11268.

3.7.2 Chemical and physical properties. Materials which support combustion or which are capable of causing an explosion shall be avoided.

3.7.3 Toxic materials. Any material which releases toxic products to the atmosphere under any condition specified herein shall not be used in the fabrication of the unit.

3.7.4 Soldering. The units shall meet the requirements of 4.11.

3.8 Nameplates and product marking. The unit marking shall be in accordance with MIL-STD-130 and MIL-M-13231 and shall include the following minimum information:

a. Nomenclature: Tray Assembly, Battery Charging MX-10154( )/U

b. ERADCOM Part Number:

c. Contract Number:

d. Manufacturer's name or FSCM"

e. Serial Number:

f. NSN

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3.9 Workmanship. The units shall meet the requirements of 4.12.

3.10 Interchangeability. (See 4.10) Interchangeability shall be in accordance with Requirement 7 of MIL-STD-454. Parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

3.11 Safety. (See 4.9) The unit shall meet the requirements of 4.13.

3.12 Human engineering. Human engineering shall meet the requirements of 4.14.

3.13 Finish. Finish color shall be approximately olive drab in accordance with color chip X24087 of FED-STD-595.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and Inspection facilities. Test and measuring equipment and facilities shall be of adequate quality and quantity to permit performance of the required examinations and tests. Unless otherwise specified herein, all examinations and tests shall be performed under prevailing temperature, humidity and atmospheric pressure conditions. The establishment and maintenance of a calibration system to control accuracy of the measuring and test equipment shall be in accordance with MIL-STD-45662.

4.1.2 Failures. If a failure occurs in any of the individual tests of the unit, the cause of the failure shall be ascertained. If the failure was caused by the malfunction of the sub-assembly or piece part it shall be corrected by replacement or repair of the failed sub-assembly or piece part. The procuring activity reserves the right to require earlier tests of the series to be repeated if deemed necessary to ensure compliance with this specification.

The deliverable unit shall have passed the individual test requirements when no failure occurs during the rerun of the tests and an engineering analysis of the specific failure indicates one of the following causes:

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a. A Random failure

b. Improper workmanship (providing repetition of poor workmanship is precluded by proper procedure).

c. An environmental limit was exceeded during testing.

4.1.3 Corrective action. The corrective action taken shall be documented and the effect of this action upon future production units as well as the effect of the lack of this corrective action on units already produced and delivered shall be defined and explained. If the unit fails the retest because of the recurrence of the identical failure or related failure attributable to the replacement or repair of the failed sub-assembly or piece part, the test in progress shall be discontinued and the deficiency analyzed and corrected to the satisfaction of the procuring activity.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Materials inspection (See 4.3)

b. First article inspection (See 4.4)

c. Quality conformance inspection of equipment before packaging (See 4.6)

d. Packaging inspection (See 4.15)

4.3 Materials inspection. Materials inspection shall consist of certification supported by verifying data that the materials listed in Table I, used in fabricating the battery tray assembly are in accordance with the applicable referenced specifications or requirements prior to such fabrication.

TABLE I. Materials inspection

Material	Requirement paragraph	Applicable Specification
Plastic molding	3.3.2.2	L-P-393
Switches	3.3.2.3	MIL-S-8805/3
Cable	3.3.2.1	MIL-C-3432 and MIL-C-3885
Heat shrink tubing	3.3.2.5	MIL-I-23053/5
Battery connectors	3.3.2.4	MIL-C-39012/21
Parts material processes	3.3	MIL-P-11268

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4.4 First article test. First article tests shall be performed as specified in the contract or purchase order, to verify that the unit meets all the requirements imposed in Section 3 of this specification. The absence of a specified requirement for testing in no way relieves the supplier of the responsibility for the delivery of a unit(s) capable of meeting all the requirements of this specification or of passing all the tests specified. In the event of design change(s), the procuring activity reserves the right to require additional testing of the first unit which reflects the design change(s).

4.4.1 First article units. The contractor shall furnish six(6) first article units of the complete Tray Assembly, Battery, Charging MX-10154( )/U.

4.4.2 First article inspection. The first article inspection shall consist of inspections specified in table II and shall be performed in the order specified therein.

4.4.3 First article data. The first article test plan and test report(s) shall be as required in the contract or purchase order.

4.5 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as a part of the inspection of equipment before packaging:

Item	Where required
Parts, materials, and processes;general	3.7
Protective coating	3.7.1
Marking	3.8
Finish, protective	3.13

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TABLE II. First article inspection.

Inspections	Required paragraph	Test paragraph	Unit 1	2	3	4	5	6
1. Inspections covered by subsidiary documents	4.3		inspections to be performed on all units					
2. Group A inspection	See Table III	4.6.1	inspections to be performed on all units					
3. Group B inspection	See Table IV	4.6.2	inspections to be performed on all units					
4. Environmental								
Temperature test	3.5.1	4.8.1	1					
Low temperature flexibility	3.5.1.1	4.8.1.3	2					
Humidity	3.5.2	4.8.2		1				
Salt fog	3.5.8	4.8.8		2				
Moisture resistance	3.5.9	4.8.9						1
Dust	3.5.3	4.8.3			1			
Vibration	3.5.5	4.8.5			2			
Shock	3.5.6	4.8.6				1		
Bench handling shock	3.5.7	4.8.7				2		
Fungus	3.5.4	4.8.4					1	

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4.6 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspection specified in 4.5, 4.6.1 through 4.6.4, and 4.7. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.6.1 Group A inspection. Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in table III. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subjected to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQLs indicated in table III.

4.6.1.1 Order of inspection within Group A. Group A inspection shall be performed in an order satisfactory to the Government.

TABLE III. Group A inspection.

Inspection	Req. Paragraph	Inspection Paragraph	AQL (dphu*) Major	AQL (dphu*) Minor
Visual and mechanical	3.3	4.3	1.0 dphu*	*
DC resistance	3.4.1	4.7.1		
Continuity	3.4.2	4.7.2	1.0% for the group	*
Insulation resistance	3.4.3	4.7.3		
Dielectric withstanding voltage	3.4.3	4.7.4		
Pull test	3.5.10	4.8.10	1.0%	
Switch function	3.4.5	4.7.5	1.0%	

* dphu - defects per hundred units.

** all electrical defects are major.

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4.6.2 Group B inspection. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to table IV and to the special inspection levels of table I, sample size code letters, of MIL-STD-105.

4.6.2.1 Group B sampling plans. The Group B sampling plans, for the AQLs listed in table IV shall be as follows:

AQL	INSPECTION LEVEL
4.0%	S-4
6.5%	S-4

TABLE IV. Group B inspection.

Inspection	Requirement Paragraph	Inspection Paragraph	AQL
Interchangeability	3.10 3.3	4.10	6.5%

4.6.3 Group C inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in table V. Samples shall be selected in accordance with 4.6.3.1.

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TABLE V. Group C inspection.

Inspection	Requirement Paragraph	Inspection Paragraph
Group 1		
Moisture resistant	3.5.9	4.8.9
DC Resistance	3.4.1	4.7.1
Continuity	3.4.2	4.7.2
Insulation Resistance	3.4.3	4.7.3
Dielectric withstanding voltage	3.4.4	4.7.4
Pull test	3.5.10	4.8.10
Switch function	3.4.5	4.7.5
Group 2		
Temperature	3.5.1	4.8.1
Low temperature flexibility	3.5.1.1	4.8.1.3
Group 3		
Dust	3.5.3	4.8.3
Humidity	3.5.2	4.8.2
Vibration	3.5.5	4.8.5
Bench handling shock	3.5.7	4.8.7
Salt fog	3.5.8	4.8.8
Group 4		
Shock	3.5.6	4.8.6
Fungus*	3.5.4	4.8.4

*The battery charging tray shall be thoroughly washed, cleaned, dried, and refurbished after this inspection before proceeding with subsequent inspection.

4.6.3.1 Sampling for Group C inspection.

4.6.3.1.1 Sampling for inspection (Group 1). One sample shall be selected at random, from every 50 equipments or fraction thereof

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produced and subjected to each inspection specified in table V. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot. (See 4.6.3)

4.6.3.1.2 Sampling for inspection (Group 2). Two samples shall be selected at random each month and subjected to each inspection in table V. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot. (See 4.6.3)

4.6.3.1.3 Sampling for inspection (Group 3). One sample shall be selected at random, from each 200 equipments produced or fraction thereof, and subjected to each inspection specified in table V. The first sample shall be selected at the start of the contract from the first quality conformance inspection production lot. (See 4.6.3)

4.6.3.1.4 Sampling for inspection (Group 4). This inspection shall be required only when First Article samples are not specified by contract or purchase order. When required, one sample shall be selected at random, at the start of the contract, from the first quality conformance inspection production lot (See 4.6.3), and subjected to each inspection specified in table V.

4.6.3.2 Order of inspection within Group C. The Group C inspections shall be performed in the order given in table V. (See 6.3)

4.6.3.3 Group C failures. Actions required relative to Group C failures shall be as specified in the contract or purchase order.

4.6.3.4 Reinspection of conforming Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract or order provided all damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.7 Electrical performance characteristics. The Tray Assembly, Battery Charging MX-10154( )/U shall be tested in accordance with 4.7.1 through 4.7.5 (See 3.4).

4.7.1 DC resistance. A direct current (dc) voltage of not more than 10 volts shall be applied across each terminated conductor so that the same current will flow through the entire length of each terminal or contact, wire and solder or crimp connection. The dc resistance shall be measured using any standard method. The accuracy of the reading shall be ±5 percent. The test shall be in accordance with Method 303 of MIL-STD-202.

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4.7.2 Continuity. Each conductor of each assembly shall be tested for continuity and correct connections between its terminations, using a potential of not more than 10v. A suitable indicator, such as a flashlight bulb, shall be placed in the circuit.

4.7.3 Insulation resistance. Insulation resistance shall be measured in accordance with Method 3003 of MIL-STD-1344. The potential shall be applied between each conductor and the shield, and metallic shells, if present. Cable connectors shall be unmated.

4.7.4 Dielectric withstanding voltage. Dielectric withstanding voltage shall be performed in accordance with Method 301 of MIL-STD-202. A dc potential of 1000 +10 volts shall be applied for one (1) minute. The potential shall be applied between each conductor or shield and the remaining conductors and shields connected together in parallel and between each conductor and metallic shell, if present.

4.7.5 Switch function test. Contractor shall conduct tests on pushbutton switches and BNC connectors to demonstrate that they function properly and that the normally closed pushbutton circuit is opened with the weight of the battery and that contact is established with the BNC connector.

4.8 Environmental condition tests. Unless otherwise specified, the service conditions shall be conducted in accordance with MIL-STD-810. Operational tests where required shall be the electrical tests specified for Group A inspection in table II.

4.8.1 Temperature tests. (See 3.5.1)

4.8.1.1 High temperature. The equipment shall be subjected to the test of Method 501, Procedure II, of MIL-STD-810. For Step 4, raise internal temperature to 160°F (71°C). For Step 7 the internal chamber temperature shall be adjusted to 145°F (63°C) for test item operation (this includes effects of solar radiation).

4.8.1.2 Low Temperature. The equipment shall be subjected to the test of Method 502, Procedure I, of MIL-STD-810. The storage temperature shall be -70°F (-57°C) and shall be maintained for a period of not less than 2 hours following stabilization of the test item. The low operating temperature shall be -25°F (-32°C).

4.8.1.3 Low-temperature flexibility. The cable assembly or portion thereof containing the molded section of heat-shrunker

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transition shall be aged in a circulating air oven for minimum of 48 hours at 160° ±2°F. The cable assembly shall then be conditioned for a minimum of 48 hours at the temperature specified in 3.5.1.1, this temperature shall be continuously recorded during conditioning and testing. Each crotch or molded section shall be held in a mechanical device so that the portion of the cable entering the molded section flexes when the cable is flexed. The cable shall be flexed manually or mechanically. Any part of a mechanical device which will be in contact with the test specimen shall be cold conditioned. At the end of the conditioning period and while at the conditioning temperature, the cable shall be flexed rapidly for two cycles through a 180-degree arc (±90 degrees from 0 degrees). The flexing procedure shall be repeated in a plane 90 degrees to the plane of initial flexing. At the conclusion of the test, the specimen shall be examined visually at room temperature under 3X magnification.

4.8.2 Humidity test. (See 3.5.2) The equipment shall be subjected to the test of Method 507, Procedure II of MIL-STD-810.

4.8.3 Dust. (See 3.5.3) The equipment shall be subjected to the test of Method 510, Procedure 1 of MIL-STD-810. The unit shall be in the operating configuration during the test.

4.8.4 Fungus. (See 3.5.4) The cable assembly model in the assembled and as ready for delivery condition, shall be subjected to 28 days of Method 508, MIL-STD-810. After 28 days, based upon visual examination, the equipment assembly shall pass if it shows no more than only sparse microbial growth with restricted tuberal growth development in an area of 1 to 10% of the total area and no more than six unrelated minute colonies with mycelial development in areas only in other than critical circuit portions; such as terminal spacing, printed circuit boards, etc., with sparse growth due to random contamination or traces of unmixed material ingredients. No operational test is required after the fungus test to pass the test. The equipment shall fail if it shows more than the growth specified above. The test shall be repeated if the control items fail to show profuse growth after 14 and 28 days of test.

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4.8.5 Vibration. (See 3.5.5) The cable assembly shall be subjected to the test of Method 514, Procedure XI, Part 2, of MIL-STD-810. The test time shall be 2 hours per axis.

4.8.6 Shock. (See 3.5.6) The equipment shall be subjected to the tests of Method 516, Procedure II of MIL-STD-810.

4.8.7 Bench handling shock. The cable assembly shall be subjected to the tests of Method 516, Procedure V of MIL-STD-810. The test curve shall be as shown in the figure for the half sine pulse configuration. The pulse level shall be 20g's and the pulse length 11 milliseconds.

4.8.8 Salt fog. (See 3.5.8) The equipment shall be subjected to the test of Method 509, Procedure I, of MIL-STD-810. Operation of the equipment is not required. Failure criteria shall be limited to corrosion, deterioration or change in tolerance limits of any internal or external parts which could in any manner prevent the test item from meeting operational service or maintenance requirements.

4.8.9 Moisture resistance. (See 3.5.9) The tray assembly shall be tested as specified in Method 106 of MIL-STD-202. The following exceptions and details shall apply:

a. Measure initial continuity and insulation [illegible].
b. No load.
c. Connectors shall be mated throughout the cycling in the chamber.
d. DC resistance and insulation resistance (see 4.3 and 4.5) measurements shall be made (without water immersion) at high humidity after Step 6 at the end of the 10th cycle, with the cable inside the humidity chamber. Assemblies with pin, lug, bare wire terminations, or other such stub ends which are not sealed shall have the cable ends brought through the chamber wall. Cable assemblies with molded strain-reliefs or seals shall have the back-end of the strain-relief or seal remain inside the chamber.

4.8.10 Pull test. (See 3.5.10) Forces shall be applied in a manner that shall avoid sudden jerks between cable components, to insure a good mechanical connection has been made between components of the finished cable assembly, terminal to cable, connector to cable, taper pin to connector, strain-relief to connector and cable, and molded bend reliefs to connectors and cable. The

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component to be tested shall be secured in a holding device to keep it stationary and the cable shall be gripped 6 to 12 inches from the component under test in such a manner that there will be no damage to the cable and no slippage in the grip. Pull forces shall be applied parallel to the axis of the cable and twist forces shall be applied in a plane perpendicular to the axis of the cable.

4.9 Visual and mechanical inspection. (See 3.3.1, 3.11, 3.13) Equipment shall be examined for the defects listed in MIL-STD-252.

4.10 Inspection for interchangeability. The dimensions listed below shall be gauged or measured to determine compliance with the physical interchangeability requirements of 3.10. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers, and insertable subassemblies and replaceable parts, when such dimensions affect mating of parts.

b. Internal dimensions of cavities in cases and assemblies, when such dimensions affect acceptability of insertable subassemblies.

c. Location of fasteners on separate parts or assemblies which must mate together such as cases, covers and mountings.

4.11 Soldering. Soldering shall be in accordance with Requirement 5 of MIL-STD-454.

4.12 Workmanship. Workmanship of the unit shall be in accordance with Requirement 9 of MIL-STD-454.

4.13 Safety. The unit shall comply with Requirement 1 of MIL-STD-454.

4.14 Human engineering. Human engineering design shall be in accordance with MIL-STD-1472.

4.15 Packaging inspection. Packaging inspection requirements specified herein are classified as follows:

a. First Article Inspection of Packaging

b. Quality Conformance Inspection of Packaging

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4.15.1 First Article Inspection of Packaging. Unless otherwise specified in the contract, First Article Inspection of Packaging shall be in accordance with the Unit Pack Design Validation Requirements of MIL-P-116.

4.15.2 Quality Conformance Inspection of Packaging.

4.15.2.1 Materials inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.15.2.2 Preservation inspection. Inspection of preservation and interior markings shall be in accordance with group A and B Quality Conformance Inspection Requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

4.15.2.3 Packing inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in table VI "PACKING INSPECTION PROVISIONS". Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and Acceptable Quality Level of 4.0 percent defective.

TABLE VI.

PACKING INSPECTION PROVISIONS.

NO.	CHARACTERISTIC	METHOD OF INSPECTION
101	Intermediate container not as specified	Visual
102	Improper closure of intermediate container	Visual
103	Shipping containers not in accordance with specification	Visual
104	Excessive cube	Visual
105	Improper blocking and bracing	Visual
106	Closure not in accordance with specification	Visual
107	Weight and size exceed container limitations	Weight & Measure
108	Strapping not in accordance with specification, incorrectly applied, omitted	Visual
109	Marking omitted, incorrect, or illegible	Visual

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5. PACKAGING

5.1 Preservation. Preservation shall be as specified in MIL-STD-726, coded as follows:

5.1.1 Level A. 1Ø-1-1-ØØ-OO-NS-X-ED-Ø-OO-A.

5.1.2 Level B 1Ø-1-1-ØØ-OO-NS-X-ED-Ø-OO-B.

5.2 Packing and marking. Packing and marking shall be in accordance with MIL-E-55585.

6. NOTES

6.1 Intended use. The MX-10154( )/U is intended to allow simultaneous charging of six(6) each Battery Assembly BB-503/TAS.

6.2 Ordering data. Procurement documents should specify:

a. Title, number and the date of this specification and and any amendment thereto.

b. Level A or B preservation and packing (See 5.1).

c. When rough handling test is required.

d. Marking and shipping requirements.

e. First article inspection.

(1) Number of units.

(2) First article test plan and test report(s) required.

f. Place of final inspection.

g. When first article inspection rough handling tests are not specified.

h. When first article packaging inspection test reports require Acquisition Activity approval prior to production unit packing.

6.3 Group C inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.6.3.2)

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6.4 Nomenclature. The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example MX-10154W/U. The contractor should apply for nomenclature in accordance with applicable clause in the contract. (See 1.1)

6.5 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.6 Failure definitions.

6.6.1 Equipment failure. The inability of the MX-10154( )/U Battery Charger Tray to perform its function as specified herein to include damage resulting from improper operation. Damage which results from abuse is not a failure nor is the need for adjustment of the operator's controls. The need for adjustment of controls not normally available to the operator is considered to be a failure. The performance requirements are not restricted to those of table II.

6.6.2 Pattern failure. The occurrence of two (2) or more failures of the same part in identical or equivalent application whose combined failure rate exceeds that predicted. When two (2) or more dependent failures of the same part occur whose combined failure rate exceeds that predicted and which are due to more than one independent failure, the dependent failures are relevant pattern failures. Unless a government approved Reliability Math Model is available failure rates found in MIL-HDBK-217 shall be used.

6.6.3 Independent failure. A failure which will independently cause equipment performance outside of specified limits - one which occurs without being related to the failure of associated items.

6.6.4 Dependent failure. A failure of a part which is a direct result of an independent failure - one which is caused by the failure of an associated item(s). Dependent failures are not necessarily present when simultaneous.

6.7 Level B preservation. When level B preservation is specified, this level of protection will only be used under known favorable conditions during transportation, storage, and handling.

Custodian:
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Preparing Activity:
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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-49331A	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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